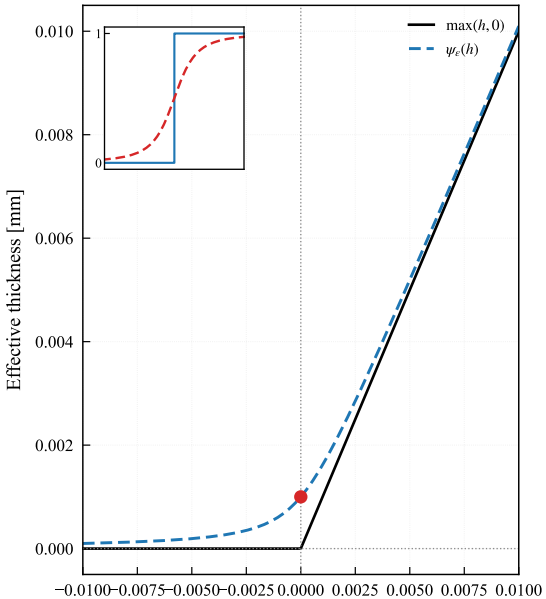
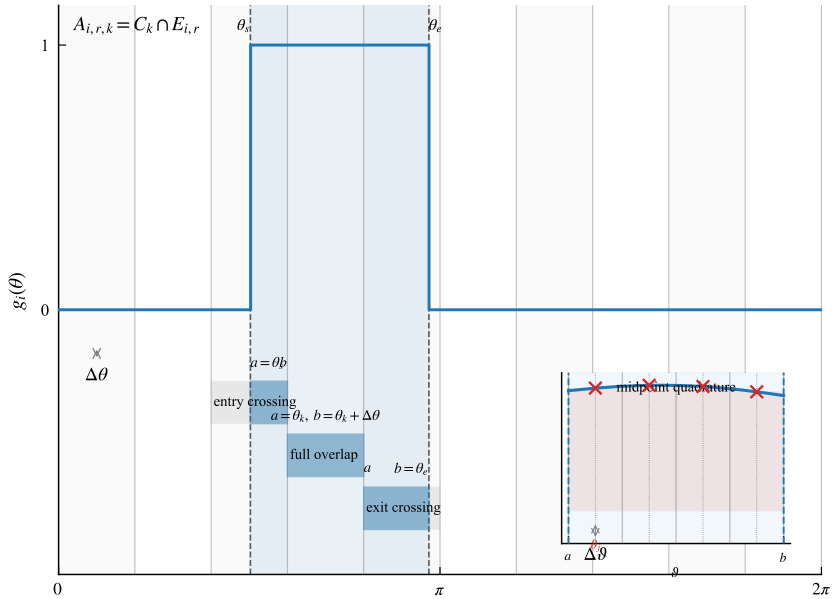

(a) Chip-loss regularisation
max(h, 0)
ψε(h)
Effective thickness [mm]
Chip thickness h [mm]
0.010
0.008
0.006
0.004
0.002
0.000
−0.0100 −0.0075 −0.0050 −0.0025 0.0000 0.0025 0.0050 0.0075 0.0100
1
0


(b) Geometric engagement and active-interval extraction
$A_{i,r,k} = C_k \cap E_{i,r}$
θ_{st}
θ_e
$g_i(\theta)$
$\Delta\theta$
$a = \theta_b$
entry crossing
$a = \theta_k,\ b = \theta_k + \Delta\theta$
full overlap
a
$b = \theta_e$
exit crossing
midpoint quadrature
a
$\Delta\theta$
θ
b
Spindle phase θ [rad]
1
0
π
2π